OSITO DRONE STEM ACADEMY

SERIES 08: ALL-ABILITIES & ADAPTIVE STEM PILOTS - AGES 5-10 (ALL ABILITIES)

THIS STEM COLORING BOOK BELONGS TO

YOUNG PILOT'S NAME / NOMBRE DEL PILOTO

MY DRONE'S CALLSIGN / NOMBRE DE MI DRON

CO-PILOT / GROWN-UP CO-PILOT NAME

Title: Vol. 353: Liam's Single-Switch Sip-and-Puff Controller at Space Rocket Imagination Lab

Featured STEM Pilot: Liam (Adaptive Wheelchair Drone Pilot & Obstacle Champion)

Flight Environment: Space Rocket Imagination Lab - Target Age: Ages 5-10 (All Abilities)

Core STEM & Math Focus: Breath Pressure Signals and Relay Counts

Includes: 8 Single-Sided Coloring Plates, 4 Vector STEM Activities, Flight Log & Certificate!

Explore Youth Drone STEM Camps, 3D Printing & Flight Simulators at ositodrone.com

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OFFICIAL YOUTH DRONE SAFETY RULES & FAA TRUST NOTICE

SAFETY RULES FOR RECREATIONAL DRONE PILOTS

1

1. ALWAYS FLY WITH A GROWN-UP

Make sure a parent, teacher, or coach supervises every flight.

2

2. FLY IN OPEN, SAFE SPACES

Stay clear of people, pets, roads, trees, and active airport airspace.

3

3. KEEP DIRECT LINE-OF-SIGHT

Always keep your quadcopter where you can see it with your own eyes.

4

4. USE DUCTED PROPELLER GUARDS

Inspect all 4 ducted prop guards and battery latch before takeoff.

5

5. COUNT & CHECK PRE-FLIGHT MATH

4 propellers + 1 charged battery + 1 adult co-pilot = Safe Takeoff!

6

6. TAKE THE FREE FAA TRUST TEST

Learn official recreational sky rules at faa.gov/uas/trust with family.

OSITO DRONE LLC – PUBLISHING & LEGAL ATTRIBUTION

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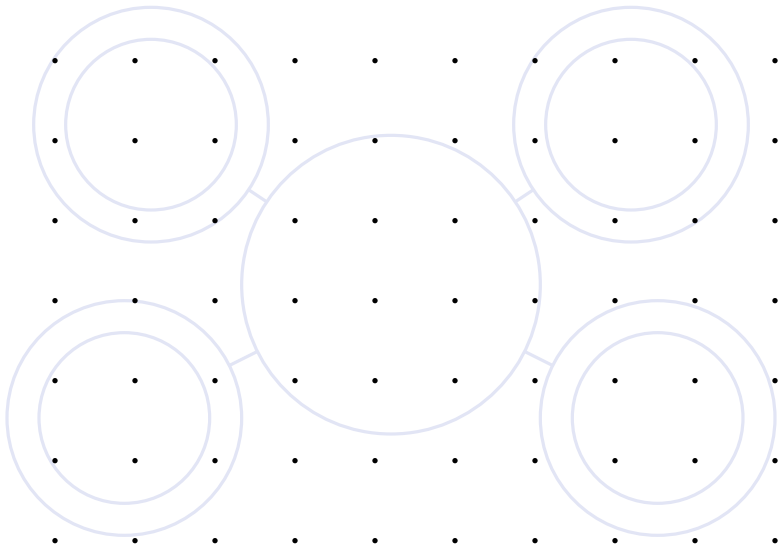
Designed to comply with FAA TRUST youth aviation educational benchmarks.

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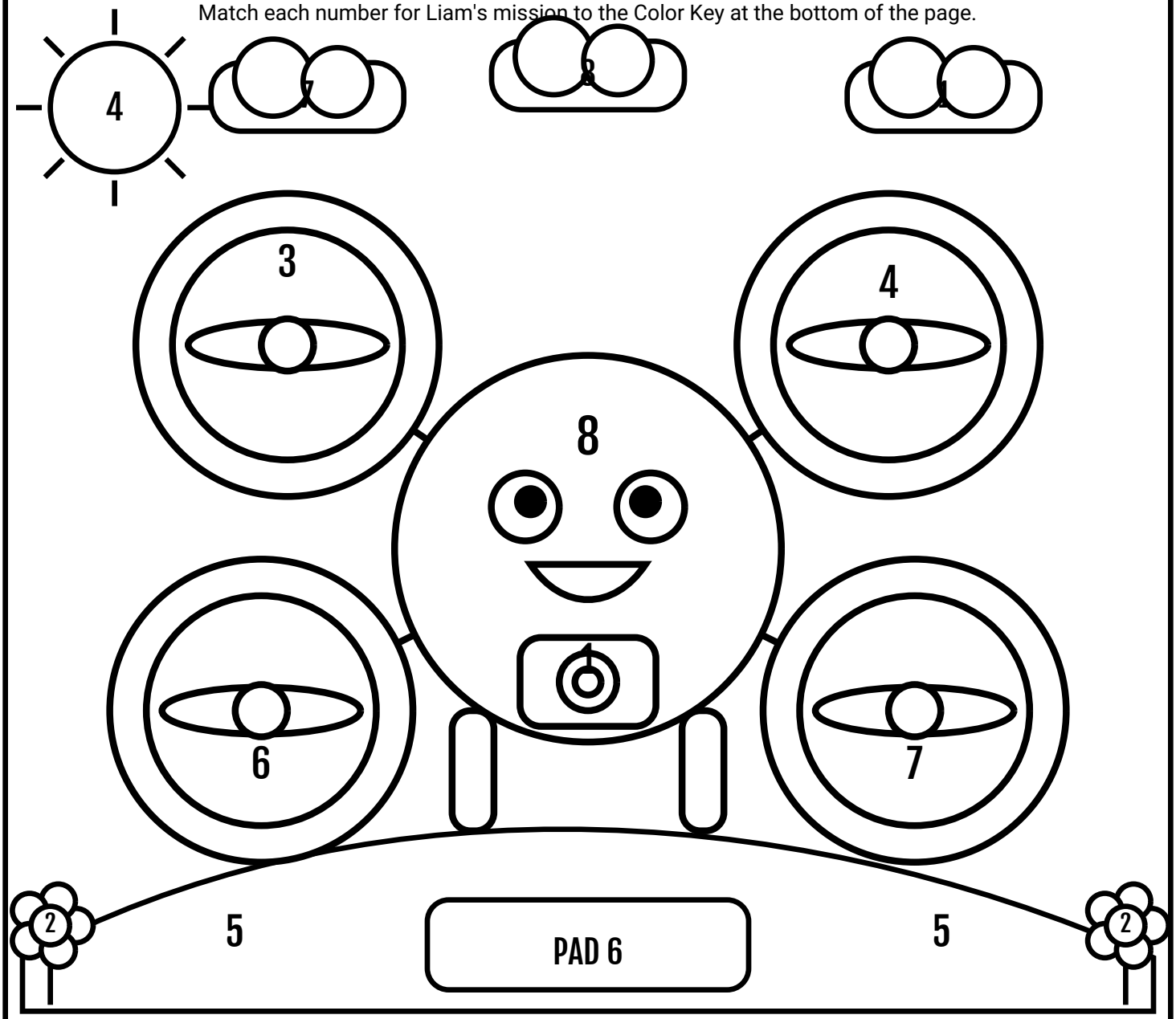


LIAM & PET DRONE – SINGLE-SWITCH SIP-AND-PUFF CONTROLLER



COLOR BY NUMBER: LIAM'S SINGLE-SWITCH SIP-AND-PUFF CONTROLLER

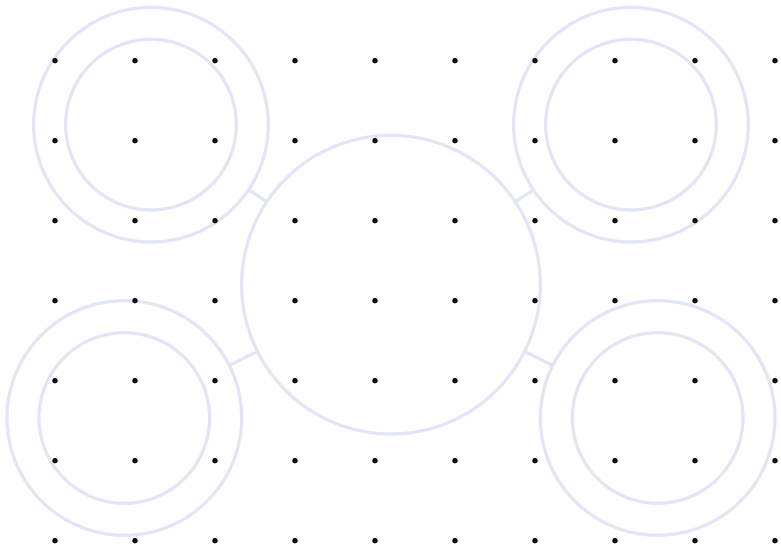
Match each number for Liam's mission to the Color Key at the bottom of the page.



NUMBER-BY-NUMBER COLOR KEY

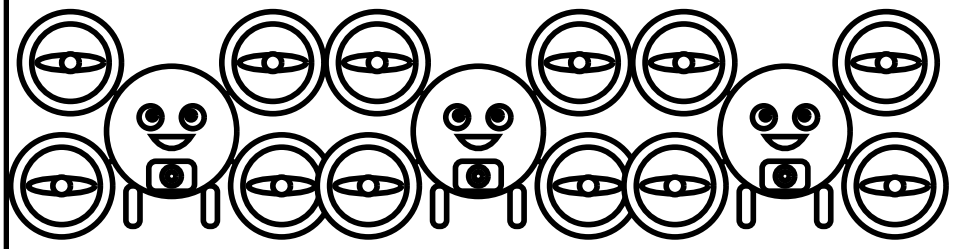
1 = RED	2 = BLUE	3 = YELLOW	4 = GREEN
5 = ORANGE	6 = PURPLE	7 = TEAL	8 = BROWN

BLEED-PREVENTION BACKING PAGE 6 - BOOK #353 - FLIGHT LOG & SKETCH PAD



DRONE MATH: LIAM'S SINGLE-SWITCH SIP-AND-PUFF CONTROLLER

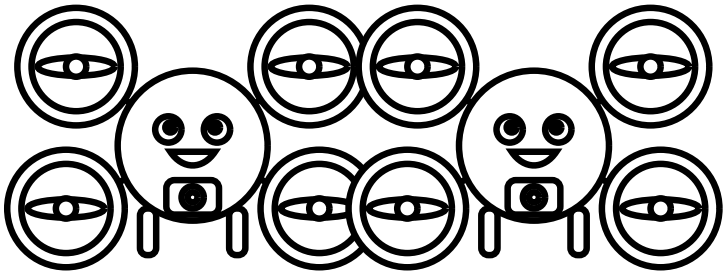
Every quadcopter has 4 ducted propellers! Solve Liam's flight problems and trace the numbers.



MISSION #1: COUNT ROTORS

3 Drones = 12 Props!

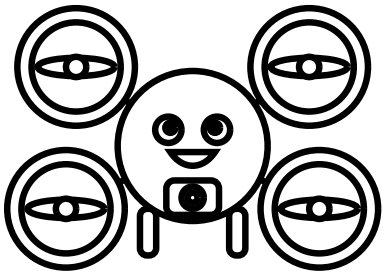
Color the drones & trace numbers!



MISSION #2: COUNT ROTORS

2 Drones = 8 Props!

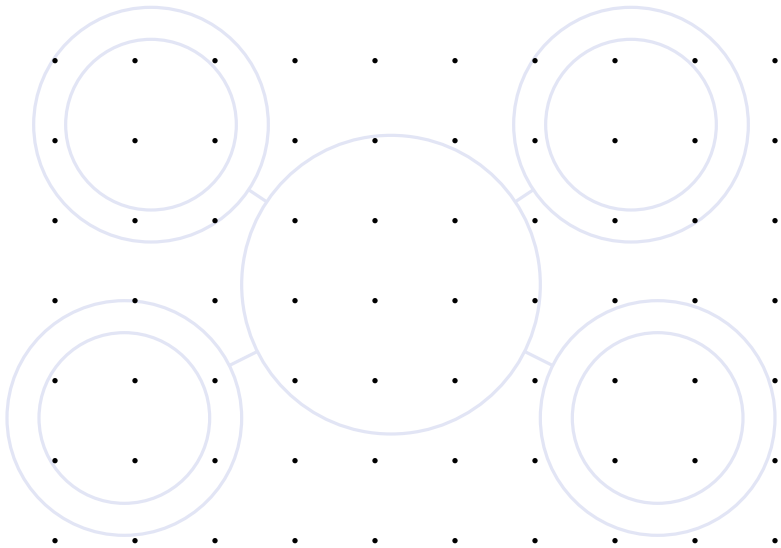
Color the drones & trace numbers!



MISSION #3: COUNT ROTORS

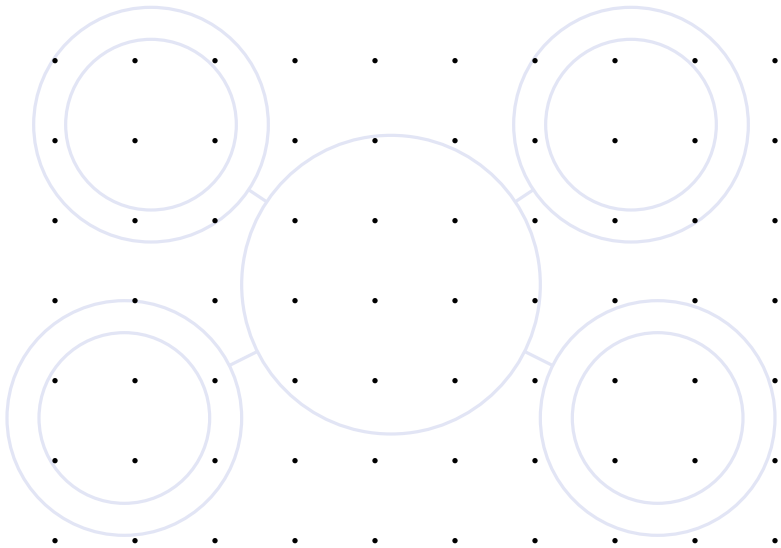
1 Drone = 4 Props!

Color the drones & trace numbers!



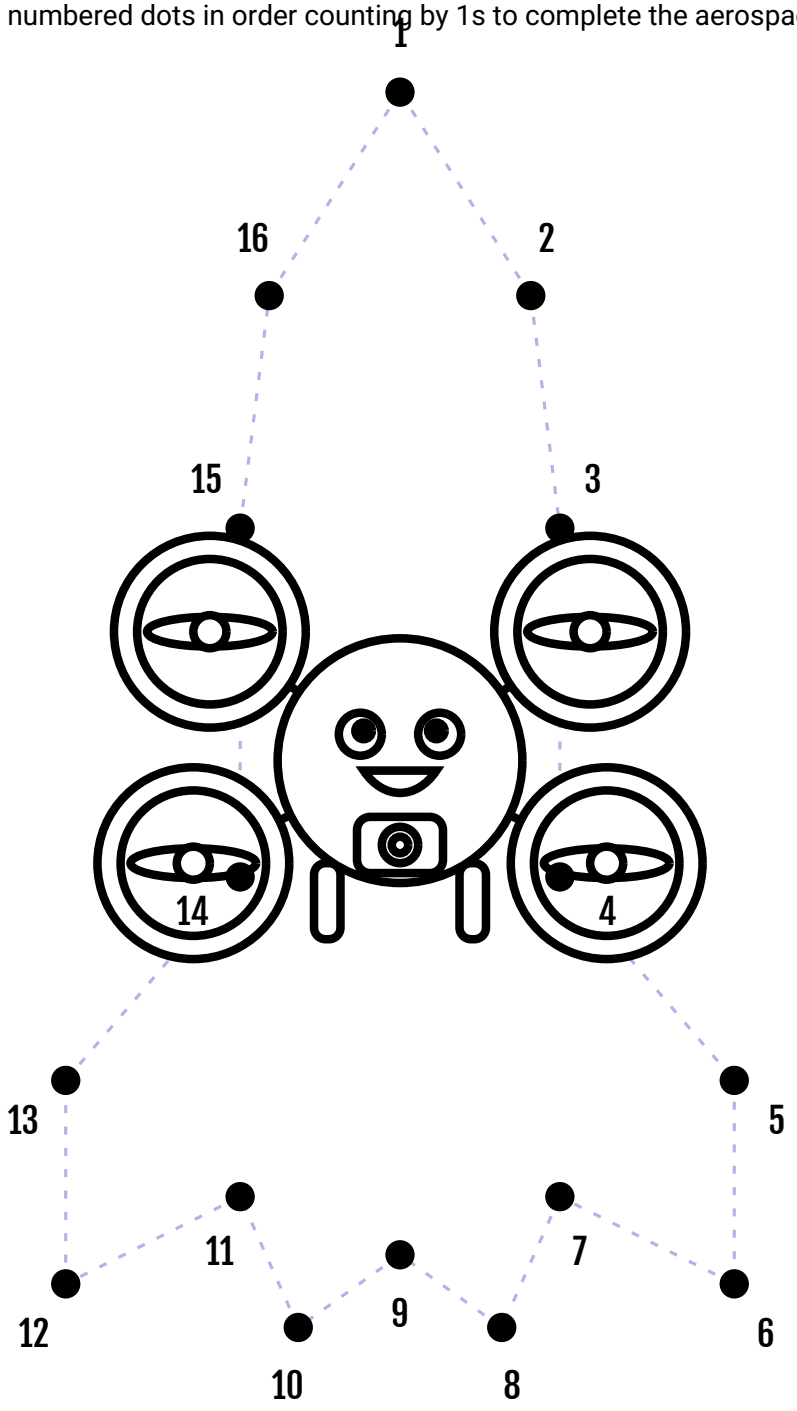


COLOR BY NUMBER & COUNT ROTORS WITH LIAM!

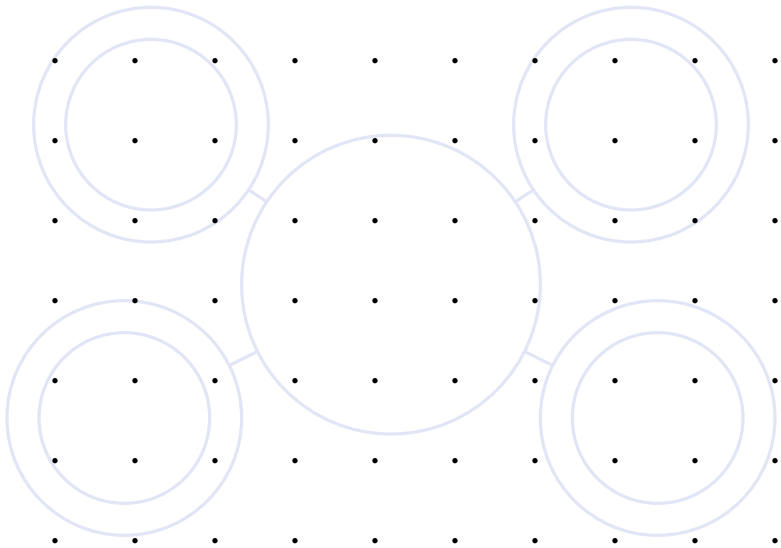


CONNECT THE DOTS: LIAM'S SINGLE-SWITCH SIP-AND-PUFF CONTROLLER

Connect the numbered dots in order counting by 1s to complete the aerospace blueprint!



BONUS FOR LIAM: You connected 16 dots counting by 1s! Final number = 16!



CODE-BY-NUMBER: SOLVE & COLOR PIXEL DRONE

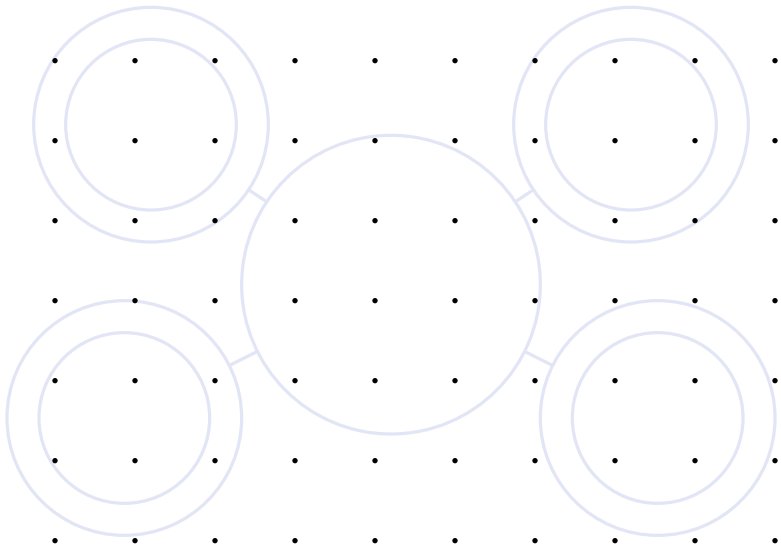
Solve or match the number in each grid square (A1-G7) using the Color Code Key below to reveal pixel art!

	1	2	3	4	5	6	7
A	5	5	5	1	5	5	5
B	5	5	1	3	1	5	5
C	5	5	2	2	2	5	5
D	5	5	2	3	2	5	5
E	5	1	2	2	2	1	5
F	1	1	3	1	3	1	1
G	4	4	4	4	4	4	4

PIXEL CODE COLOR KEY

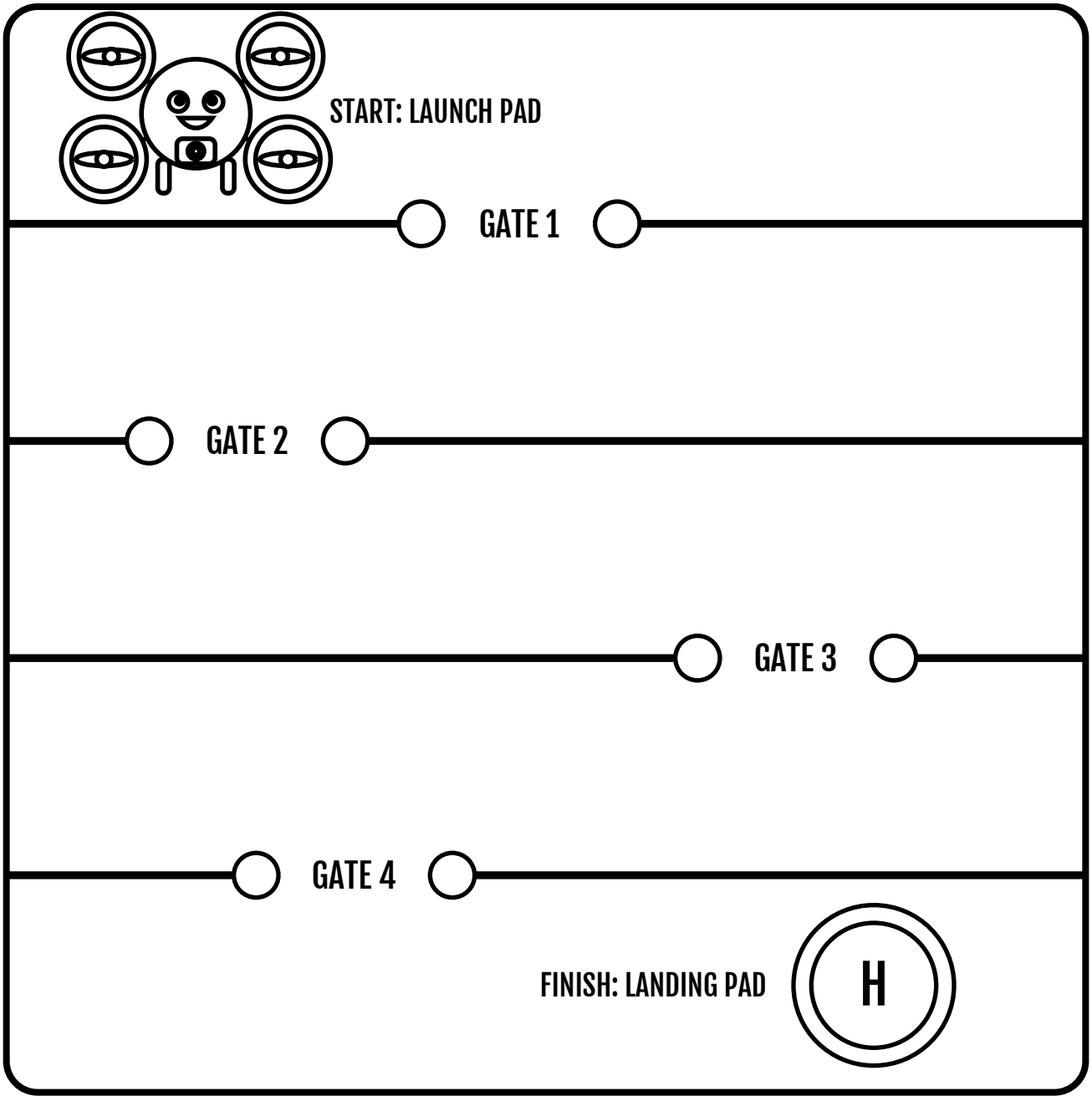
1 = RED 2 = BLUE / INDIGO 3 = GOLD / YELLOW 4 = GREEN 5 = LIGHT SKY BLUE

STEM Tip: Computers and drone cameras store pictures as grids of numbered pixels just like this!



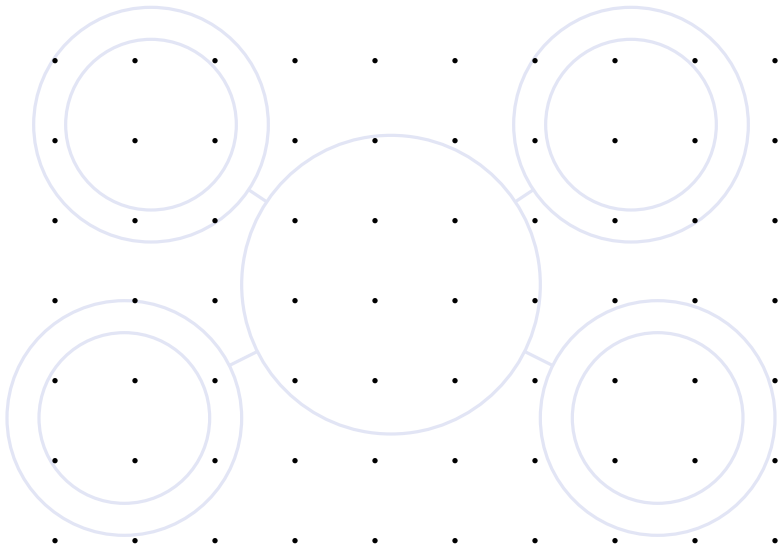
SAFE FLIGHT-PATH MAZE: GUIDE LIAM'S DRONE!

Draw a line from LAUNCH at top-left through open gates in order (1 -- 2 -- 3 -- 4) to LANDING PAD 'H'!



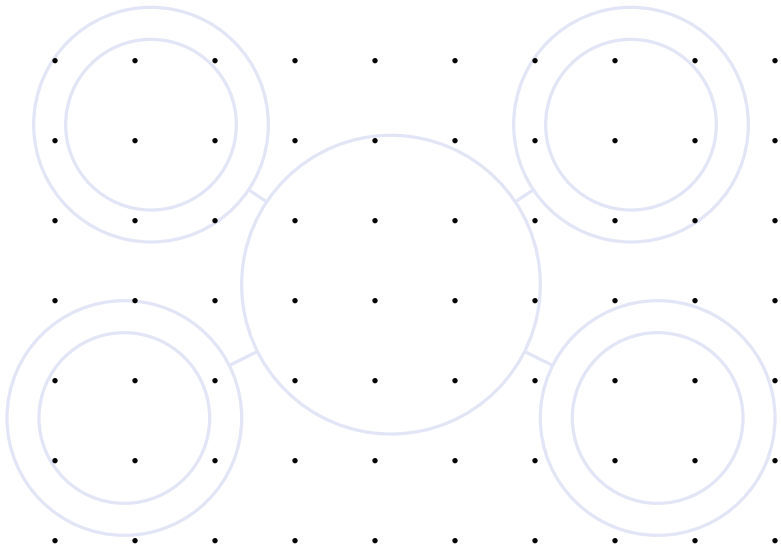
COLORING CHALLENGE: Color all 4 Gate Hoops different colors and draw arrows along your safe flight path!

BLEED-PREVENTION BACKING PAGE 16 - BOOK #353 - FLIGHT LOG & SKETCH PAD



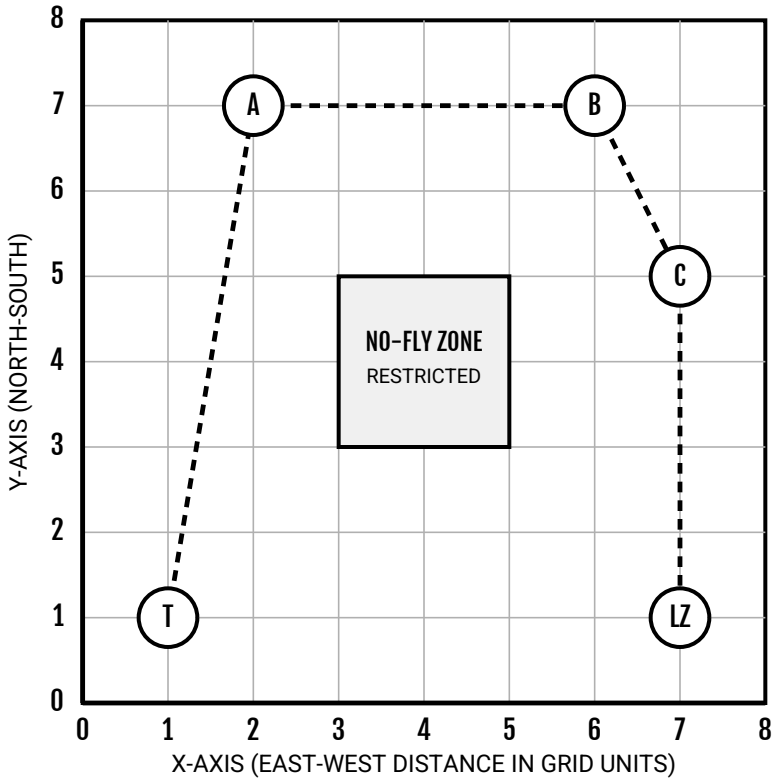


SAFE DRONE FLIGHT AT SPACE ROCKET IMAGINATION LAB



COORDINATE FLIGHT PLAN: LIAM'S WAYPOINT NAVIGATION

Plot waypoints on the (X, Y) grid for Liam's Single-Switch Sip-and-Puff Controller!

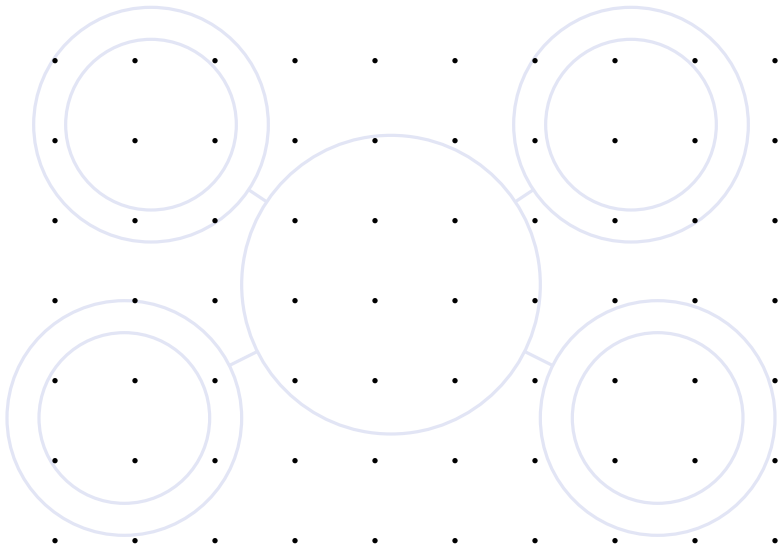


FLIGHT PLAN WAYPOINTS

WP T:	Takeoff Pad	(1 , 1)
WP A:	Liam Survey	(2 , 7)
WP B:	Waypoint Single-Switc	(6 , 7)
WP C:	Bridge Check	(7 , 5)
WP LZ:	Landing Zone	(7 , 1)

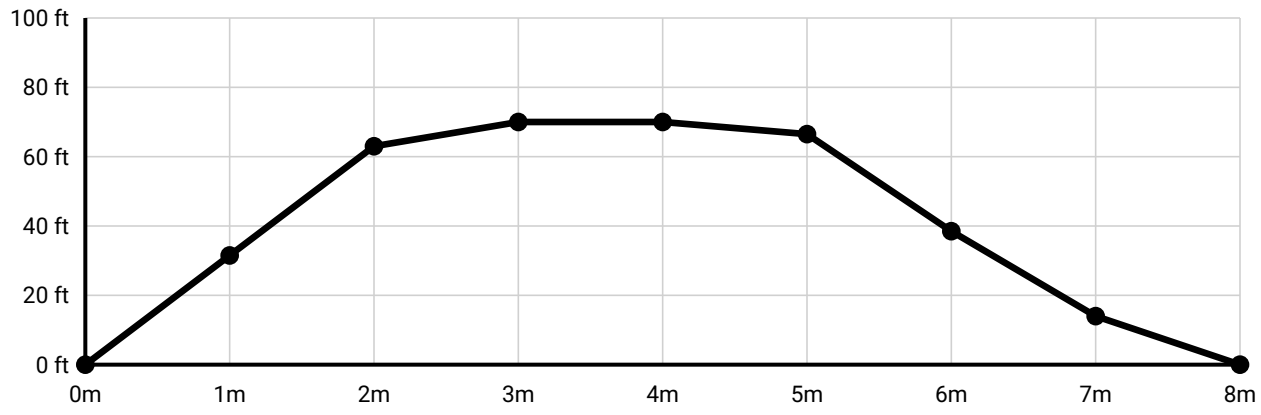
COORDINATE CHALLENGE:

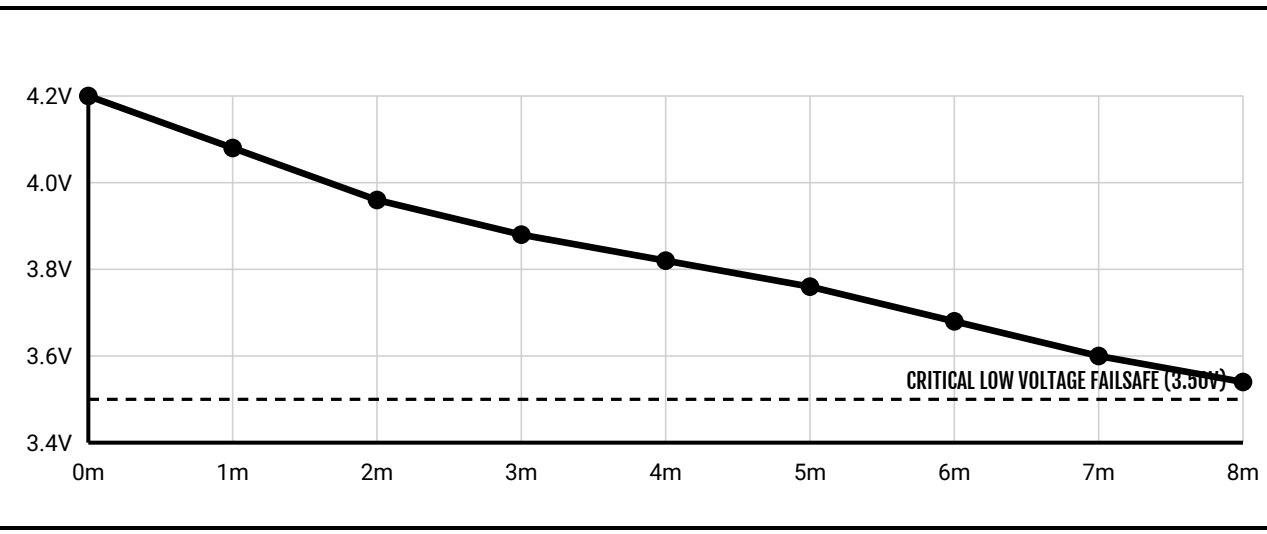
1. Waypoint B coordinates: (6 , 7)
2. Did flight avoid No-Fly Zone? [X] YES
3. Total flight legs flown: 4 legs

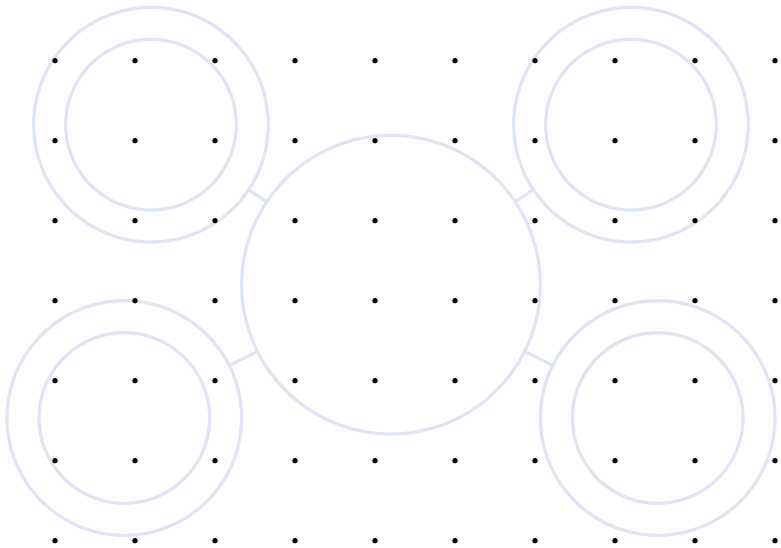


FLIGHT TELEMETRY: ALTITUDE & BATTERY DISCHARGE ENVELOPE

Analyze real flight telemetry data downloaded from the onboard BlackBox flight recorder.

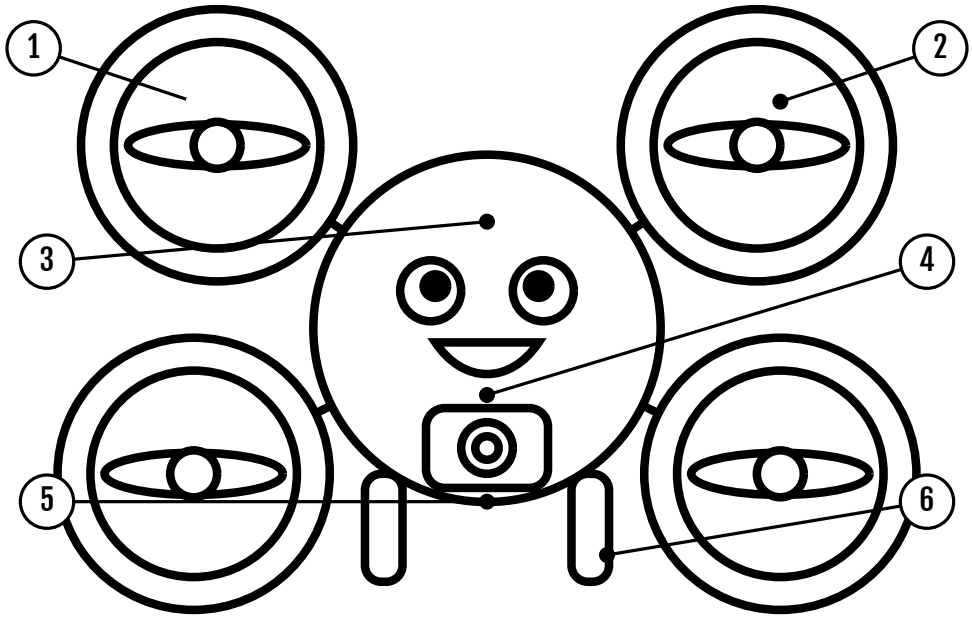






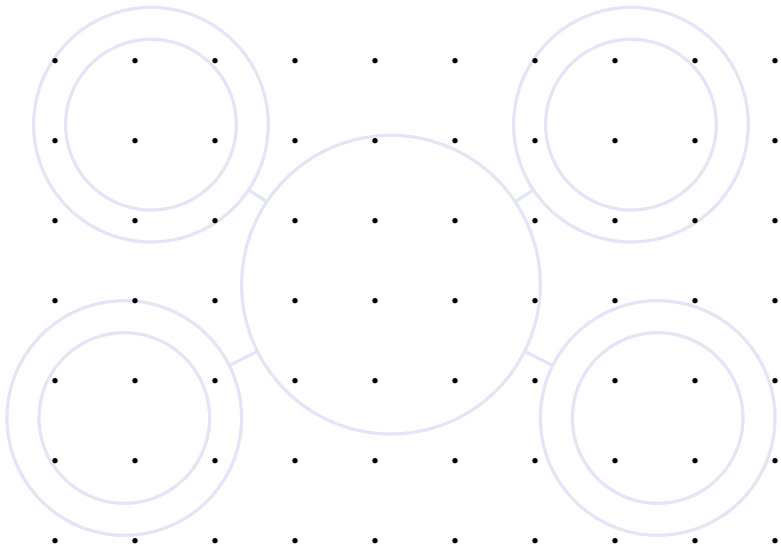
AEROSPACE BLUEPRINT: CUB-75 AVIONICS & AIRFRAME ANATOMY

Study the mechanical airframe and avionics architecture of the Osito CUB-75 STEM training drone.



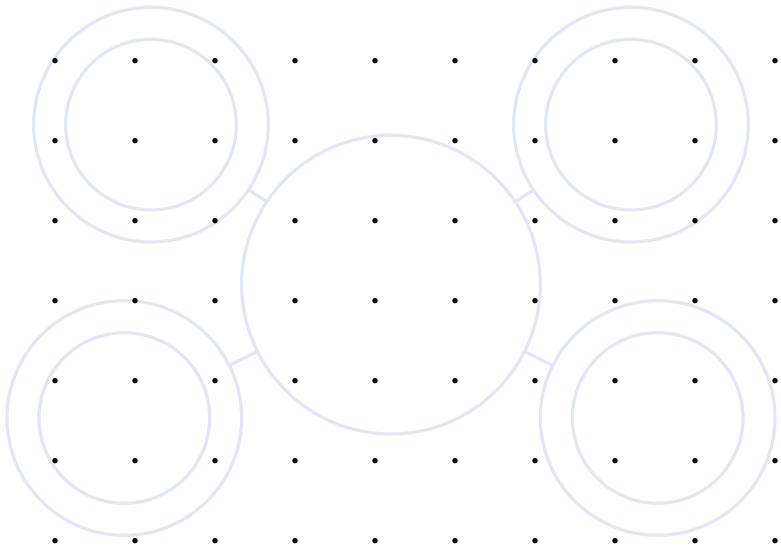
1	**Ducted Prop Guards**	Duct rings that shield propellers and provide 360-degree impact safety.
2	**Brushless DC Motors**	0802 19000KV motors that spin props at 30,000 RPM to produce vertical lift.
3	**Flight Controller (FC)**	Microprocessor & gyroscope that balances the drone 1,000 times every second.
4	**4-in-1 ESC Board**	Electronic Speed Controller regulating power sent from battery to each motor.
5	**Micro FPV Camera**	Ultra-wide lens sensor broadcasting real-time video to pilot FPV monitors.
6	**1S LiPo Battery**	3.8V High-Voltage Lithium Polymer rechargeable cell providing flight energy.

CUB-75 SPECS: Wheelbase: 75mm - AUW Weight: 39.5g - Motors: 0802 19000KV - Flight Time: 8.0 min



FLIGHT PHYSICS: THRUST-TO-WEIGHT & BATTERY ENDURANCE MATH

Apply real aeronautical engineering equations to calculate multirotor performance and endurance.



OFFICIAL OSITO JUNIOR PILOT FLIGHT MISSION LOG

RECORD YOUR ACTUAL FLIGHT MISSIONS AND COMPLETED STEM CHALLENGES

FLIGHT #	DATE	FLIGHT DURATION	MISSION OBJECTIVE	COACH SIGNATURE
Mission 1				
Mission 2				
Mission 3				

OSITO STEM PILOT CERTIFICATE OF ACHIEVEMENT

THIS CERTIFIES THAT JUNIOR PILOT

PILOT NAME HERE

HAS COMPLETED ALL STEM ACTIVITIES AND SAFETY PROTOCOLS IN

BOOK #353: SINGLE-SWITCH SIP-AND-PUFF CONTROLLER

Core Aeronautical Skill: Breath Pressure Signals and Relay Counts

AND IS OFFICIALLY RECOGNIZED AS A CERTIFIED JUNIOR OSITO DRONE PILOT!

JUNIOR PILOT SIGNATURE

COACH / PARENT SIGNATURE & DATE

